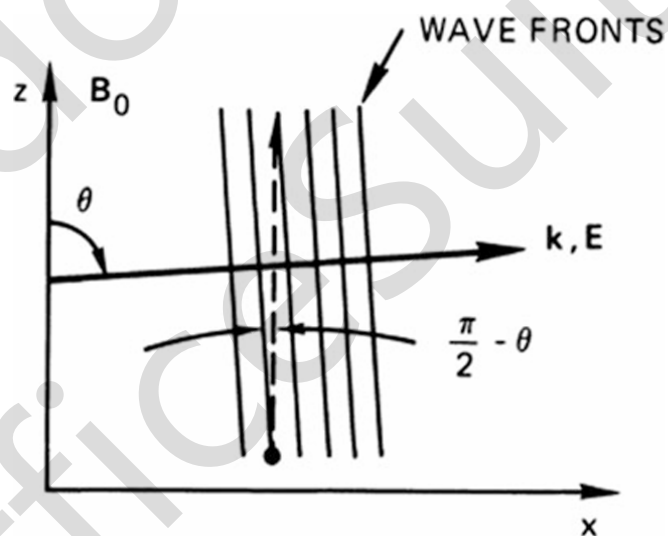


Variables approach;

Electrostatic Ion Waves Perpendicular to B We next consider what happens to the ion acoustic wave when k is perpendicular to B_0 . It is tempting to set $k \cdot B_0$ exactly equal to zero, but this would lead to a result (Sect. 4.11) which, though mathematically correct, does not describe what usually happens in real plasmas. Instead, we shall let k be almost perpendicular to B_0 ; what we mean by "almost" will be made clear later. We shall assume the usual infinite plasma in equilibrium, with n_0 and B_0 constant and uniform and $v_0 \ll E_0 \ll 0$. For simplicity, we shall take $T_i \ll 0$; we shall not miss any important effects because we know that acoustic waves still exist if $T_i \ll 0$. We also assume electrostatic waves with $k \cdot E \neq 0$, so that $E \neq \nabla \phi$. The geometry is shown in Fig. 4.23. The angle $\frac{1}{2}\pi - \theta$ is taken to be so small that we may take $E \approx E_1 \hat{x}$ and $\nabla \approx ik \hat{x}$ as far as the ion motion is concerned. For the electrons, however, it makes a great deal of difference whether $\frac{1}{2}\pi - \theta$ is zero, or small but finite. The electrons have such small Larmor radii that they cannot move in the x direction to preserve charge neutrality; all that the E field does is make them drift back and forth in the y direction. If θ is not exactly $\pi/2$, however, the electrons can move along the dashed line (along B_0) in Fig. 4.23 to carry charge from negative to positive regions in the wave and carry out Debye shielding. The ions cannot do this effectively because their inertia prevents them from moving such long distances in a wave period; this is why we can neglect kz for ions. The critical angle $\chi \approx \frac{1}{2}\pi - \theta$ is proportional to the ratio of ion to electron parallel velocities: $\chi \approx (m/M)^{1/2}$.

Plot the ellipse for $\omega_p/\omega_c = 1, 2$, and 1 . (e) Show that if $\omega_c > \omega_p$, the lower root for ω is always less than ω_p for any $\theta > 0$ and the upper root always lies between ω_c and ω_h ; and that if $\omega_p > \omega_c$, the lower root lies below ω_c while the upper root is between ω_p and ω_h .

Fig. 4.23 Geometry of an electrostatic ion cyclotron wave propagating nearly at right angles to B_0

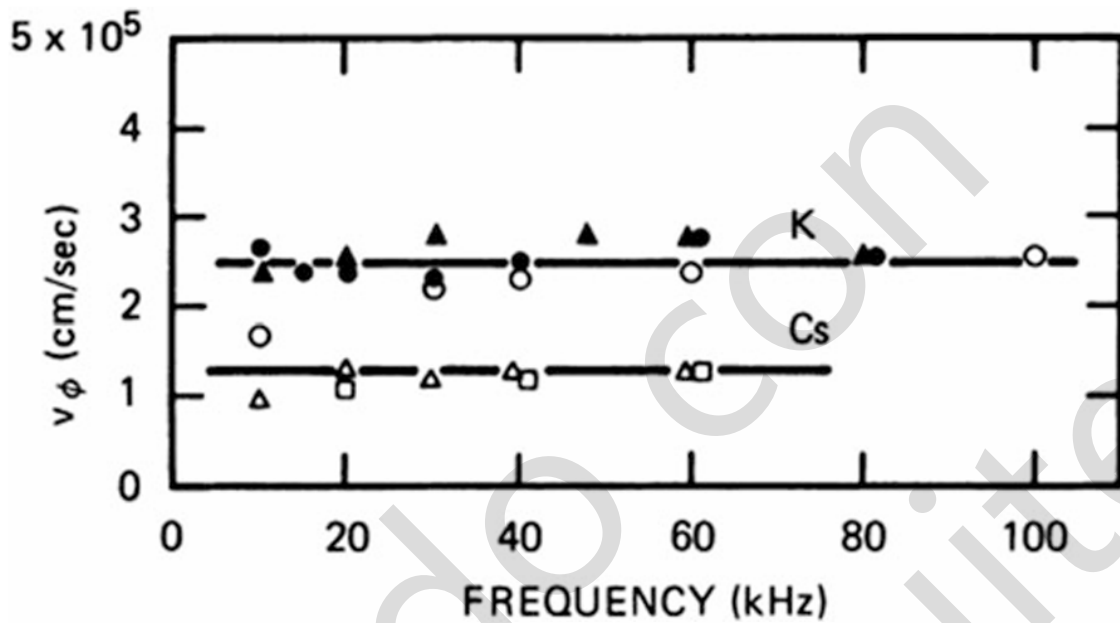


The Trivelpiece Gould dispersion curves for electrostatic electron waves in a conducting cylinder filled with a uniform plasma and a coaxial magnetic field. [From A. W. Trivelpiece and R. W. Gould, J. Appl. Phys. 30, 1784 (1959).]

For the upper hybrid oscillation, show that the elliptical orbits (Fig. 4.19) are always elongated in the direction of k . (Hint: From the equation of motion, derive an expression for v_x/v_y in terms of ω/ω_c .) 4.9 Find the dispersion relation for electrostatic electron waves propagating at an arbitrary angle θ relative to B_0 .

wave is transverse, B_1 is finite, and the wave is electromagnetic. It is of course possible for k to be at an arbitrary angle to B_0 or E_1 ; then one would have a mixture of the principal modes presented here. Coming back to the

electron oscillations perpendicular to B_0 , we shall assume that the ions are too massive to move at the frequencies involved and form a fixed, uniform background of positive charge. We shall also neglect thermal motions and set $kT_e \gg 0$. The equilibrium plasma, as usual, has constant and uniform n_0 and B_0 and zero E_0 and v_0 . The motion of electrons is then governed by the following linearized equations



Measured phase velocity of ion waves in potassium and cesium plasmas as a function of frequency. The different sets of points correspond to different plasma densities. [From Wong, Motley, and D'Angelo, loc. cit.]

The constancy of u_s with ω and n_0 is demonstrated experimentally, and the two sets of points for K and Cs plasmas show the proper dependence on M .

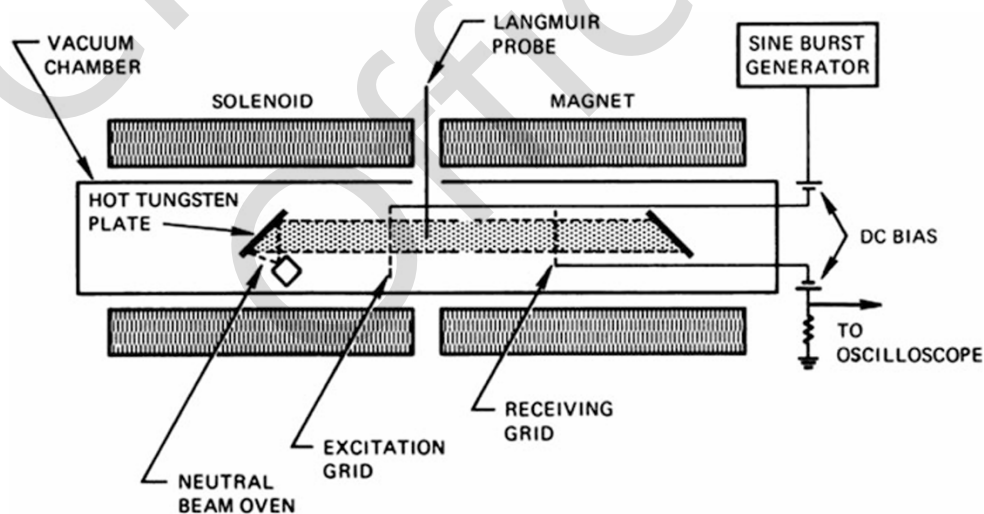
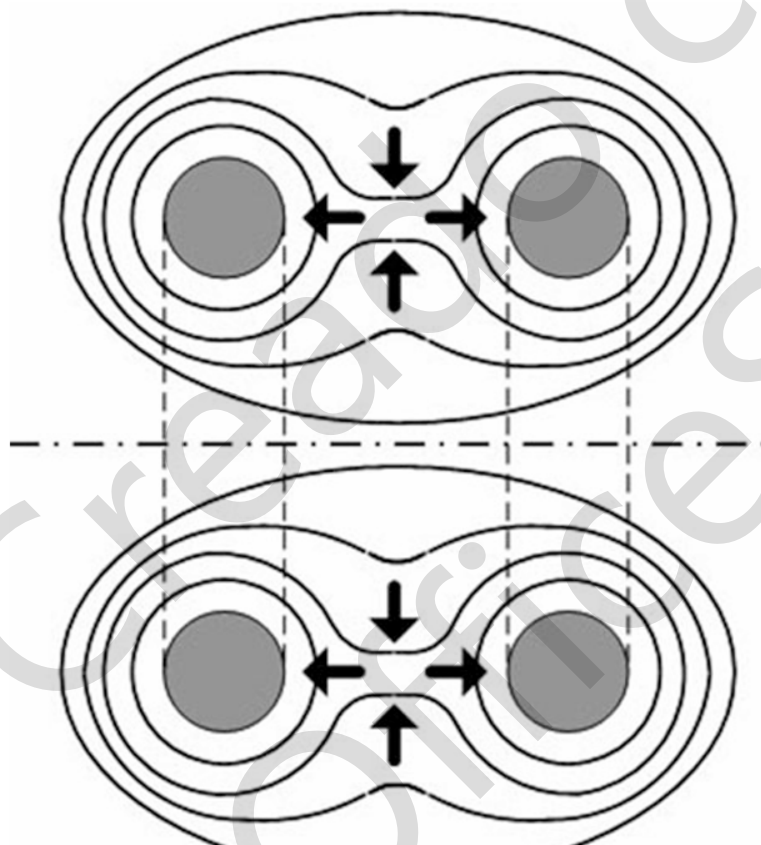


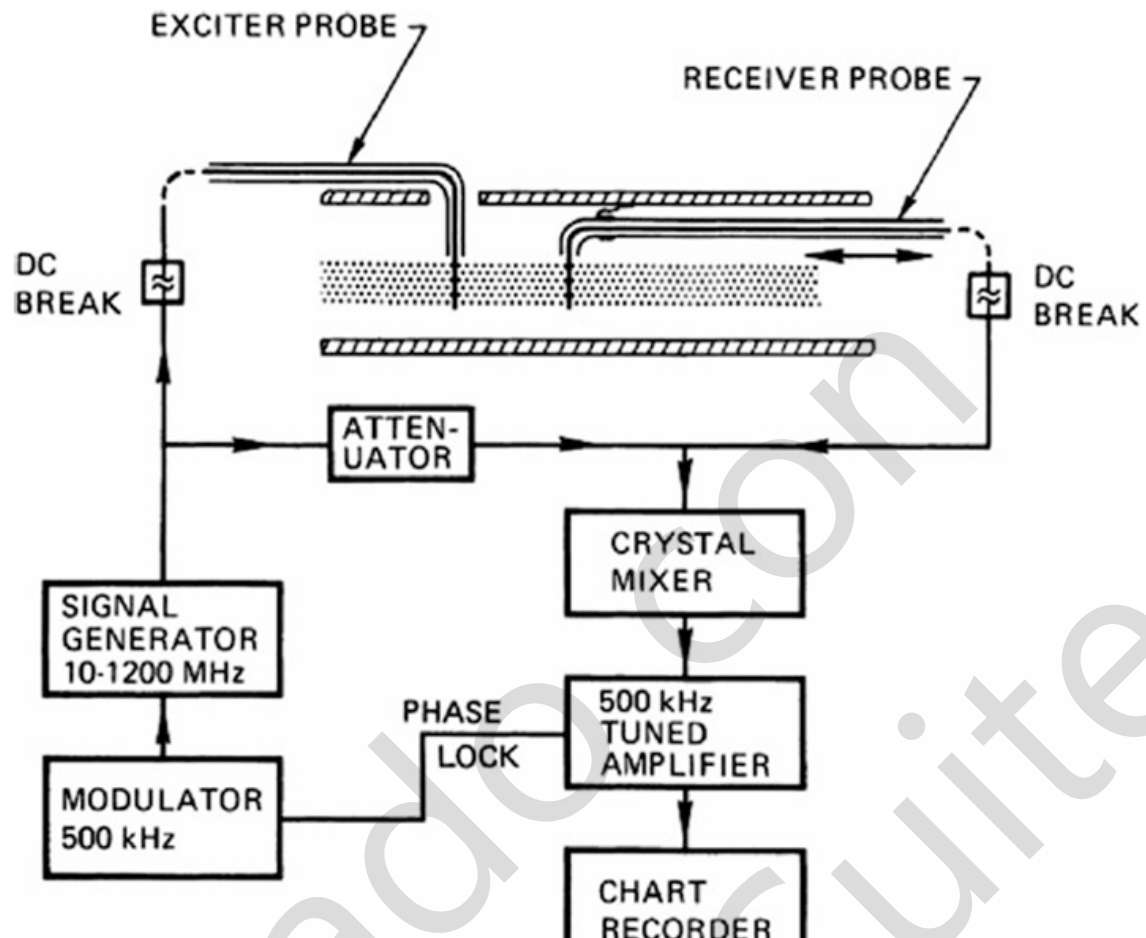
Fig. 4.14 Q-machine experiment to detect ion waves. [From N. Rynn and N. D'Angelo, *Rev. Sci. Instrum.* **31**, 1326 (1960).]

the limit $T_i \rightarrow 0$. Here Ω_p is the ion plasma frequency. For high frequencies (short wavelengths) the ion acoustic wave turns into a constant-frequency wave. There is thus a complementary behavior between electron plasma waves and ion acoustic waves: the former are basically constant frequency, but become constant velocity at large k ; the latter are basically constant velocity, but become constant frequency at large k . This comparison is shown graphically in Fig. 4.13. Experimental verification of the existence of ion waves was first accomplished by Wong, Motley, and D'Angelo. Figure 4.14 shows their apparatus, which was again a Q-machine.

dispersion curve for ion waves (Fig. 4.12) has a fundamentally different character from that for electron waves (Fig. 4.5). Plasma oscillations are basically constant-frequency waves, with a correction due to thermal motions. Ion waves are basically constant-velocity waves and exist only when there are thermal motions. For ion waves, the group velocity is equal to the phase velocity. The reasons for this difference can be seen from the following description of the physical mechanisms involved. In electron plasma oscillations, the other species (namely, ions) remains essentially fixed. In ion acoustic waves, the other species (namely, electrons) is far from fixed; in fact, electrons are pulled along with the ions and tend to shield out electric fields arising from the bunching of ions.



Neglecting viscosity, we can write the Navier–Stokes equation (3.48), which describes these waves, as $\rho \frac{\partial \mathbf{v}}{\partial t} = -\nabla p - \frac{1}{4} \nabla p^2 - \gamma p \nabla \rho$



Electrons streaming into adjacent layers of plasma with their thermal velocities will carry information about what is happening in the oscillating region. The plasma oscillation can then properly be called a plasma wave. We can easily treat this effect by adding a term ∇p_e to the equation of motion:

Points: The plasma density in the lower ionosphere has been measured during satellite re-entry to be about 10^{18} m³ at 50 km altitude, 10^{17} at 70 km, and 10^{14} at 85 km.

What are the plasma frequencies there? 4.3 Calculate the plasma frequency with the ion motions included, thus justifying our assumption that the ions are essentially fixed. (Hint: include the term $n_1 i$ in Poisson's equation and use the ion equations of motion and continuity.)

4.4 For a simple plasma oscillation with fixed ions and a space-time behavior $\exp[i(kx - \omega t)]$, calculate the phase δ for ϕ_1 , E_1 , and v_1 if the phase of n_1 is zero. Illustrate the relative phases by drawing sine waves representing n_1 , ϕ_1 , E_1 , and v_1 (a) as a function of x at $t=0$, (b) as a function of t at $x=0$ for $\omega/k > 0$, and (c) as a function of t at $x=0$ for $\omega/k < 0$.

The gradient in jB causes the Larmor radius to be larger at the bottom of the orbit than at the top, and this should lead to a drift, in opposite directions for ions and electrons, perpendicular to both B and ∇B . The drift velocity should obviously be proportional to r_L/L and to $v_\perp$. Consider the Lorentz force $F = qv \times B$, averaged over a gyration. Clearly, $F_x \neq 0$; since the particle spends as much time moving up as down. We wish to calculate F_y ; in an approximate fashion, by using the undisturbed orbit of the particle to find the average.

$$\mathbf{E} + \mathbf{v} \times \mathbf{B} = 0$$

(2.13)

